

SHUBHAM KISHOR KALE

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AI Engineer building LLM document-processing systems, agentic RAG with published evaluations, and the data platforms they run on.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript

Concepts & Methods: Generative AI (RAG, LLM Agents, MCP, Fine-tuning), Agentic AI, Structured Outputs, LLM Evaluation, LLM Guardrails, Vector Search, Prompt Engineering, Machine Learning, Deep Learning, NLP, Feature Engineering, Causal Inference, REST APIs, Server-Sent Events (SSE), ETL, Data Modeling

Frameworks, Tools & Cloud: PyTorch, scikit-learn, XGBoost, Hugging Face, Claude API, Gemini API, Groq API, LangChain, LangGraph, FastMCP, ChromaDB, RAGAS, FastAPI, Next.js, React, Streamlit, Supabase, Docker, Dagster, Apache Airflow, AWS (S3, Redshift), Snowflake, PostgreSQL, Pydantic, pytest, GitHub Actions (CI/CD), Claude Code, Codex

PROJECTS

AI Auditor · [GitHub] · [Live Demo] · *LLM Document Extraction, Structured Outputs, Guardrails, Full-Stack*

- Deployed a 5-stage audit preparation workspace (FastAPI, React, Supabase) where a vision LLM extracts document facts and proposes judgments while deterministic Python owns materiality, sampling, and approval, so the model never signs off.
- Implemented structured extraction over PDFs, scans, and spreadsheets through a strict JSON-schema contract with a Groq-hosted vision model, screened by 16 deterministic, policy-cited evidence checks before any fact reaches a workpaper.
- Enforced audit controls in code: row-level security across 7 Postgres tables, row-locked versioned approvals, a staleness cascade that invalidates downstream work on any correction, and a chat assistant that can propose edits but not approve; 62 tests.

AI Financial Research Agent · [GitHub] · [Live Demo] · *Full-Stack GenAI, Agentic RAG, MCP, SSE, Evaluation*

- Architected an agentic RAG system over SEC 10-K filings (5 companies, 67K+ chunks) with LangChain, LangGraph ReAct agents, a FastMCP server, ChromaDB, and Gemini, with tools discovered at runtime via MCP.
- Built and deployed a Next.js and TypeScript chat UI streaming source-grounded answers with live citations via Server-Sent Events over a two-service FastAPI and MCP backend; GitHub Actions CI/CD runs lint, eval, and 85 tests.
- Benchmarked the agent on 66 labeled questions across 6 types with RAGAS and a cross-family judge check, scoring empty and recursion-limit answers as zero; a model swap lifted faithfulness 0.71 to 0.88 and cut terminal failures from 12 to 6.

WORK EXPERIENCE

Quantegy Analytics | New York, USA

May 2025 - Present

Analytics Engineer

- Shipped an LLM-powered navigation assistant (Claude API) into a production analytics platform serving 50+ daily users across 5 markets, routing natural-language queries to the right dashboard via a structured page-catalog context.
- Ranked retail endcap placement across 420 stores for a consumer products client using a stacked decision-tree and linear-regression ensemble over engineered demographic features; installed stores posted a 12% first-week sales lift (\$118K) against 0.2% in matched control stores.
- Adopted AI-assisted development with Claude Code as daily practice on client-shipping work, migrating 5 executive dashboards to the .pbip text format so agent-authored changes ship under Git-based code review.
- Engineered Dagster-orchestrated ELT pipelines ingesting 4M+ patient and financial records from 15+ Redshift tables into Snowflake, powering analytics across 3 healthcare markets.
- Scheduled 10+ table ingestion on Airflow (largest table 10M+ records) with pytest/httpx tests on mocked APIs.
- Modeled the target Snowflake schemas behind 3 client data platforms: fact and dimension tables with conformed dimensions, declared grain, and Type 2 history tracking across raw vendor EHR, API, and CSV feeds.
- Owned data engineering and delivery for a correctional healthcare client: Dagster pipelines ingesting 120+ vendor EHR and CSV tables into Snowflake across 15 counties, plus a production Streamlit app with per-county, per-role access control.
- Delivered the underlying Streamlit analytics platform for a healthcare client, cutting a 3-day Excel reporting cycle to same-day.

Mettler-Toledo | Mumbai, India

Mar 2024 - Jun 2024

Data Analyst Intern

- Designed an end-to-end RAG system with fine-tuned T5/BART models (Hugging Face) for automated KPI commentary, with vector search over 10M+ financial records (85%+ relevance); removed 15 hrs/month of manual write-ups.

Marcellus Investment Managers | Mumbai, India

Mar 2023 - Nov 2023

Quantitative Analyst Intern

- Developed a regression-based impact-cost model ($R^2 = 0.82$) for trade-execution optimization and a forensic-screening ML ensemble (84% accuracy, 87% specificity) adopted into ongoing fraud-risk screening.

EDUCATION

Stony Brook University, New York, USA | *Master of Science in Data Science* | **GPA: 3.78/4**

Aug 2024 - Dec 2025

NMIMS University, Mumbai, India | *Bachelor of Technology in Data Science* | **CGPA: 3.56/4**

Jul 2019 - May 2023